

ED405: TIME SERIES ECONOMETRICS

**TIME SERIES ANALYSIS OF
ELECTRICITY GENERATION IN INDIA
(2019–2025)**

**INTEGRATED MASTER OF SCIENCE
IN
QUANTITATIVE ECONOMICS AND DATA SCIENCE**

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Time Series Analysis of Electricity Generation in India (2019–2025)

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November 10, 2025

Abstract

This report presents a time series analysis of India’s monthly electricity generation data from 2019 to 2025. The dataset, sourced from Ember Energy, was aggregated across all states to obtain national-level generation figures in gigawatt-hours (GWh). The objective was to identify underlying patterns, test for stationarity, and forecast future electricity generation using SARIMA modeling. Results indicate a strong upward trend and annual seasonality, with the SARIMA(1,1,1)(0,1,1)₁₂ model providing the best fit for short-term forecasting.

1 Introduction

Electricity generation is a fundamental indicator of economic growth and energy demand. India, being one of the fastest-growing economies, has experienced a steady increase in power generation capacity over recent years. Understanding the temporal dynamics of electricity generation can assist policymakers and energy planners in forecasting future demand and optimizing production strategies. This study applies time series methods to model and forecast monthly electricity generation in India.

2 Data Description

The dataset was obtained from the *Ember Energy India Electricity Data Portal* and contains monthly observations of power generation from January 2019 to 2025. Each record represents total generation (in GWh) aggregated across all Indian states.

Statistic	Value (GWh)
Mean	262,947
Standard Deviation	35,664
Minimum	182,999
Maximum	334,829

Table 1: Summary statistics of India’s monthly electricity generation.

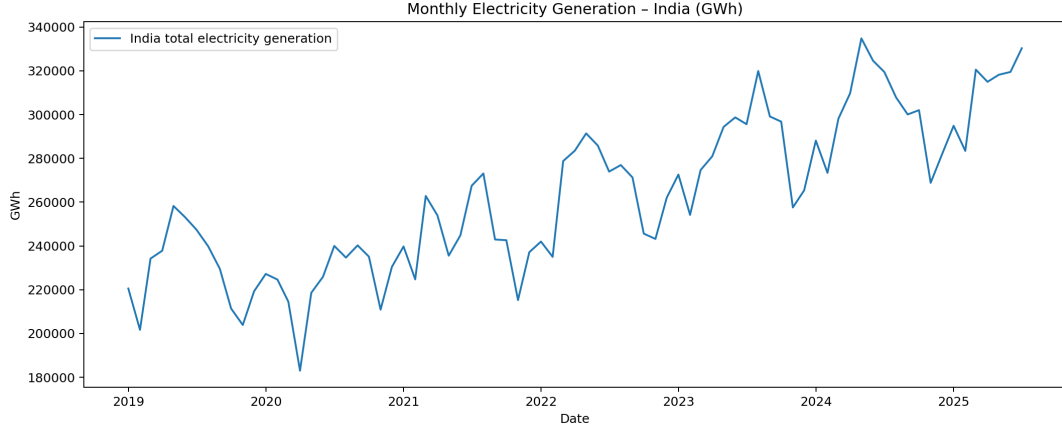


Figure 1: Monthly total electricity generation in India (2019–2025).

3 Methodology

The methodology follows four main steps:

1. **Decomposition:** The series was decomposed into trend, seasonal, and residual components using additive seasonal decomposition with a 12-month period.
2. **Stationarity Testing:** The Augmented Dickey–Fuller (ADF) test was applied. The original series had a p-value of 0.9785, indicating non-stationarity. After first differencing, the p-value dropped below 0.001, confirming stationarity.
3. **Modeling:** Various SARIMA models were tested, and the best model was selected based on the lowest AIC. The optimal configuration was SARIMA(1,1,1)(0,1,1)₁₂.
4. **Diagnostics:** Model residuals were evaluated for independence, normality, and constant variance using standardized residual plots, Q-Q plots, and correlograms.

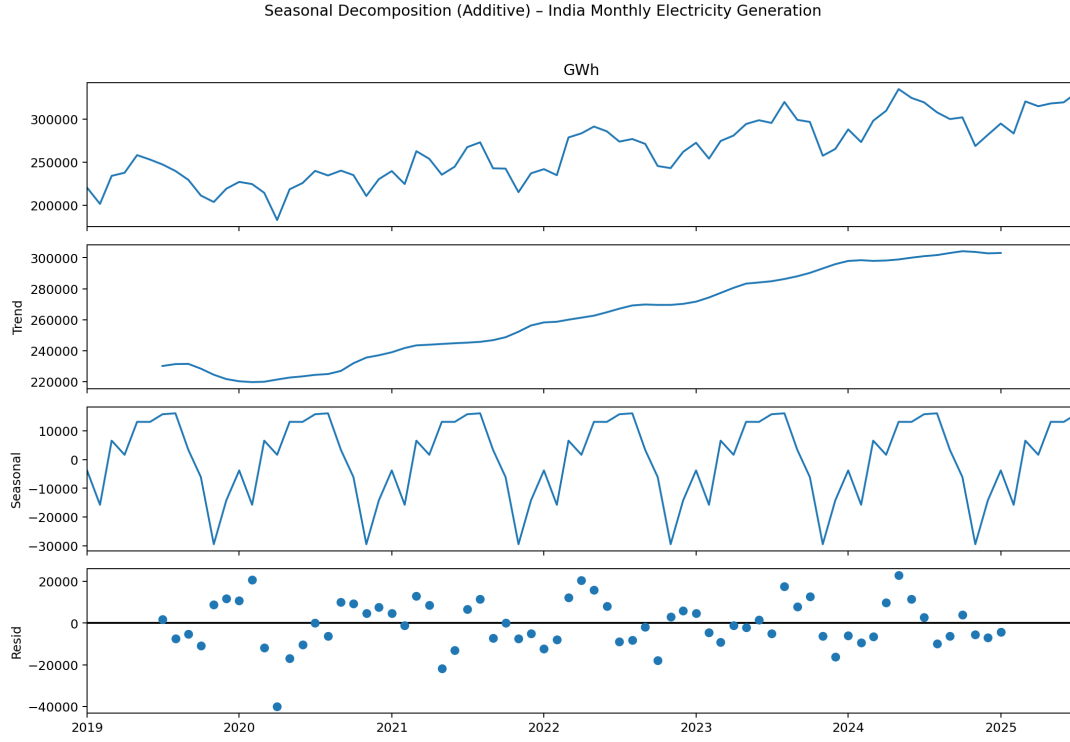


Figure 2: Seasonal decomposition of India’s monthly electricity generation.

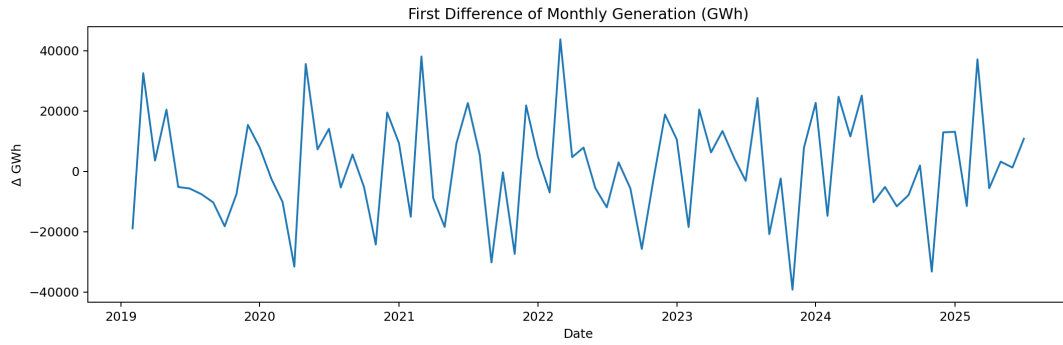


Figure 3: First difference of monthly generation showing stationarity around zero mean.

4 Results and Discussion

The decomposition revealed a consistent upward trend and clear annual seasonality, with generation peaking during the summer months and dipping during the monsoon. The SARIMA(1,1,1)(0,1,1)₁₂ model provided the lowest AIC value (1152.88).

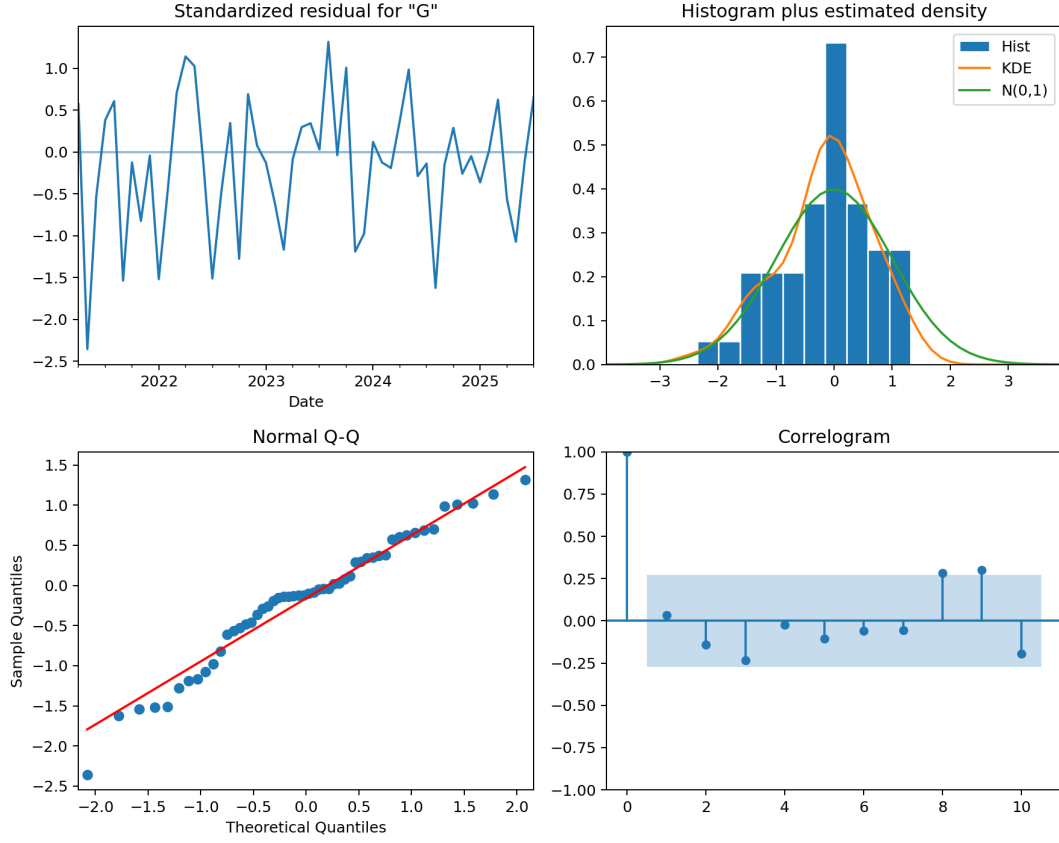


Figure 4: Model diagnostic plots: residuals appear normally distributed and uncorrelated.

Residual diagnostics confirmed model adequacy:

- Residuals are centered around zero and show no trend (white noise).
- The Q-Q plot follows a straight line, confirming approximate normality.
- The correlogram shows no significant autocorrelation, validating the model's assumptions.

The 12-month forecast indicates a continued upward trajectory in total generation, consistent with increasing energy demand.

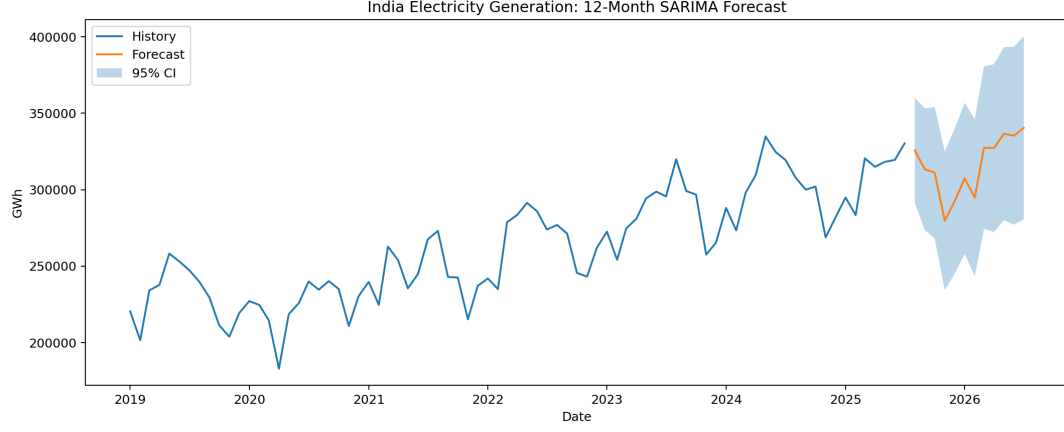


Figure 5: 12-month SARIMA forecast of electricity generation in India.

5 Conclusion

The analysis demonstrates that India's electricity generation has been steadily increasing with strong seasonal variation. The $SARIMA(1,1,1)(0,1,1)_{12}$ model successfully captures these dynamics and provides a reliable short-term forecast. Future extensions could explore the impact of exogenous factors such as temperature, GDP, or renewable penetration using SARIMAX or VAR models.

References

1. Ember Energy (2025). *India Electricity Data Portal*. <https://ember-energy.org/data/india-electricity-data/>